Andrea López López

PhD Candidate, Aerospace Engineering Sciences
Ann and H.J. Smead Department of Aerospace Engineering Sciences
University of Colorado Boulder

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Education

Ph.D., Aerospace Engineering Sciences Aug. 2021 – expected Spring/Summer 2027
University of Colorado Boulder
Advisors: Prof. Hanspeter Schaub, Prof. Jay W. McMahon
Dissertation: *Observability-Informed Adaptive Gaussian Mixture Filtering for Cislunar Angles-Only Relative Navigation*
Comprehensive exam passed March 2026.

M.S., Aerospace Engineering Sciences 2024
University of Colorado Boulder

M.S., Aerospace Engineering (Ingénieur Civil en Aérospatiale), *Magna Cum Laude* 2019
Université de Liège, Belgium (TIME double-degree program)
Thesis: *Rotation Curve of the Spiral Galaxy NGC 7331 from the Luminosity and Mass Profiles*

M.S., Aeronautical Engineering 2019
Universidad Politécnica de Madrid, Spain
Specialization: Space Vehicles

B.S., Aerospace Engineering 2016
Universidad Politécnica de Madrid, Spain
Specialization: Aerospace Science and Technology

Research Appointments

Graduate Research Assistant Aug. 2021 – present
Autonomous Vehicle Systems (AVS) Laboratory and Orbital Research Cluster for Celestial Applications (ORCCA) Laboratory, University of Colorado Boulder

- Cislunar angles-only relative navigation: nonlinear observability analysis, Gaussian mixture filtering (GMSRUKF), CR3BP relative dynamics.
- Touchless electrostatic potential sensing of spacecraft using X-ray and electron-based methods; wide field-of-view binary X-ray sensing for space object detection.

Young Graduate Trainee, Quality, Dependability & Safety Jun. 2019 – Jun. 2021
European Space Agency (ESA-ESTEC), Noordwijk, Netherlands

- Performed technical risk assessments in the Concurrent Design Facility; improved risk management methods and tooling (VBA, SQL).
- Supported RAMS analyses and verification activities across ESA projects (PDR, CDR, QR, AR).

Publications

Author name in bold. Reverse chronological within category.

Peer-Reviewed Journal Articles

- **A. López**, J. W. McMahon, H. Schaub, "Square Root Filters for Cislunar Angles-Only Relative Orbit Estimation," *Journal of the Astronautical Sciences*, 2026. **(In Review)**
- **A. López**, H. Schaub, "Detecting Space Objects with Binary Wide Field-of-View X-Ray Sensing," *Journal of Spacecraft and Rockets*, 2025. **(In Review)**
- J. Hammerl, **A. López**, Á. Romero Calvo, H. Schaub, "Touchless Potential Sensing of Differentially-Charged Spacecraft Using X-Rays," *Journal of Spacecraft and Rockets*, 2023. doi:10.2514/1.A35492

Peer-Reviewed Conference Papers

- **A. López**, J. Kulik, J. W. McMahon, "Observability Considerations for Cislunar Angles-Only Relative Navigation," AMOS Conference, 2026. **(In Preparation)**
- **A. López**, J. W. McMahon, H. Schaub, "Gaussian Mixture Square Root Filters for Cislunar Angles-Only Relative Orbit Determination," AAS/AIAA Astrodynamics Specialist Conference, Boston, MA, Aug. 2025.
- **A. López**, H. Schaub, "Relative Orbit Estimation with Wide Field of View Binary X-ray Sensing," AMOS Conference, 2023.
- J. Hammerl, **A. López**, H. Schaub, "Electric Potential Estimation of Inhomogeneous and Differentially Charged Objects Using X-Rays," AIAA SciTech Forum, National Harbor, MD, Jan. 2023. doi:10.2514/6.2023-1398
- **A. López**, J. Hammerl, H. Schaub, "Dynamic Detection of Nearby Space Objects with Binary Wide Field of View X-Ray Sensing," AIAA SciTech Forum, National Harbor, MD, Jan. 2023. doi:10.2514/6.2023-1397
- **A. López**, J. Hammerl, H. Schaub, "Detecting Space Objects With Binary Wide Field of View X-Ray Sensing," AAS Astrodynamics Specialist Conference, Charlotte, NC, Aug. 2022. Paper AAS 22-602.
- J. Hammerl, **A. López**, Á. Romero Calvo, H. Schaub, "Measuring Multiple Potentials of a Rotating and Differentially-Charged Object Simultaneously Using X-Rays," 16th Spacecraft Charging Technology Conference, Apr. 2022.
- J. Hammerl, Á. Romero Calvo, **A. López**, H. Schaub, "Touchless Potential Sensing of Complex Differentially-Charged Shapes Using X-Rays," AIAA SciTech Forum, San Diego, CA, Jan. 2022. doi:10.2514/6.2022-2312
- Carnicero Domínguez et al. (incl. **A. López**), "GEOBS: Medium Spatial Resolution Optical Instrument Concept for Disaster and Security Monitoring from GEO," Proc. SPIE 11530, Sep. 2020. doi:10.1117/12.2573805

Conference Presentations (Presentation Only)

- **A. López**, J. W. McMahon, H. Schaub, "Parametric Investigation into Orbit Stability of Small Trojan Binaries," Binaries in the Solar System VI, 2025.
- **A. López**, H. Schaub, "Touchless Time-Varying Electrostatic Potential Sensing in Orbit: Fusion of X-ray and Electron Methods," Spacecraft Charging and Technology Conference, 2024.
- **A. López**, H. Schaub, "Object Relative Heading Estimation with Binary Wide Field of View X-Ray

Sensing," Applied Space Environments Conference, 2023.

Awards and Fellowships

NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST)
2023 – 2026
Grant No. 80NSSC23K1623

Amelia Earhart Fellowship, Zonta International 2024

Teaching and Mentoring

Graduate Teaching Assistant, ASEN 6014 "Formation Flying" Fall 2025
University of Colorado Boulder. Graded all homework and projects; held weekly office hours.

Undergraduate Research Mentor, Discovery Learning Apprentice (DLA) Program 2024 – 2025
University of Colorado Boulder. Mentored a senior undergraduate in graduate-level research practice: problem decomposition, literature review, and technical communication.

Service and Leadership

Chair, Aerospace Graduate Student Organization (AGSO) 2023 – 2025
Smead Aerospace, University of Colorado Boulder. Led the organization for two consecutive years; previously served as co-chair and first-year representative. Coordinated social, professional development, mentoring, and outreach programming; advocated for graduate student interests in departmental decision-making.

Member, Smead Aerospace Graduate Committee ∼2022 – 2025
Graduate student representative on the departmental committee for ∼3 years.

Co-Founder and Organizer, CCAR Student Seminar Series Sep. 2022 – present
Colorado Center for Astrodynamics Research, University of Colorado Boulder. Co-founded and continue to organize a biweekly student seminar series, hosting 4–5 seminars per semester with two student presenters each.

Professional Memberships

- American Astronautical Society (AAS), Student Member

Technical Skills and Languages

Programming: Python, MATLAB.
Languages: Galician and Spanish (native); English (professional); French (intermediate, B2/C1, not in recent use).